

Arun Sabaratnam

🌐 Website | ✉ asaba059@uottawa.ca | 🔗 LinkedIn

Experience

Scotiabank

Jan 2026 - Present

Software Engineer Intern

Toronto, ON

- Architected a Java Spring Boot API and Python Cloud Function pipeline with OAuth 2.0 and mTLS to automate CSV-to-**BigQuery** ingestion, processing **700K+** user records with atomic MERGE operations.
- Built a GCS-to-BigQuery metadata pipeline with MD5 checksum validation and variance thresholds, ensuring data integrity across multiple tables and reducing governance overhead.
- Built a **RAG**-powered chatbot with FastAPI, Firecrawl, and GPT-4o, enabling employees to query internal knowledge bases via vector search and a multi-provider LLM abstraction layer.

Statistics Canada

May 2025 - Sep 2025

AI Software Developer Intern

Ottawa, ON

- Modernized legacy Python workflows by building high-performance **PySpark** pipelines on Microsoft Fabric, optimizing processing via strategic shuffling, partitioning, and caching.
- Trained a FastText NLP model for business categorization, improving accuracy by **20%** and using MLFlow for model versioning and registry management.
- Automated data pipelines to ingest and consolidate CRA tax records, enabling downstream analytics and reporting in the MS Fabric ecosystem.

Department of National Defence

Sep 2024 - Jan 2025

Full-Stack Developer Intern

Ottawa, ON

- Streamlined forecasting tools by refactoring database schemas with Alembic and SQLAlchemy, managing migrations that reduced query latency by **30%**.
- Revamped data-intensive UIs in Dash based on user feedback sessions, delivering a **30%** improvement in page load speeds and administrative workflow efficiency.
- Strengthened reliability by reaching **95%+** code coverage with Pytest and upgrading CI/CD pipelines, cutting production defects by **30%**.

MindBridge AI

Jan 2024 - May 2024

Data Analyst Intern

Ottawa, ON

- Automated ingestion workflows with Pandas and NumPy to process **50+** weekly financial datasets, reducing client onboarding wait times by **3x**.

Payments Canada

May 2023 - Sep 2023

Software Developer Intern

Ottawa, ON

- Implemented a serverless Google Cloud Function to automate employee probation updates via the Dayforce API, adding OAuth 2.0 auth and robust error handling to cut runtime from **45 mins to 2 mins**.
- Built a Python automation to enforce Google Drive naming conventions across **300+** files, implementing duplicate detection and bulk renaming for audit-ready organization and retention compliance.

Projects

What-The-Tech | TypeScript, Next.js, PostgreSQL, Gemini AI, Claude AI

Jan 2026

- Built an AI-powered SaaS platform using TypeScript, Next.js, and Supabase (PostgreSQL) that analyzes GitHub repositories and delivers insights through Google Gemini chat with **RAG** and Claude 3.5-generated Mermaid architecture diagrams.

Education

University of Ottawa

Expected May 2026

Bachelor of Applied Science in Software Engineering

Ottawa, ON

Technical Skills

Languages: Python, Java, JavaScript, SQL, MSSQL, C++, C#, Scala, Dart, Swift, HTML/CSS

Frameworks/Libraries: Spring Boot, Node.js, FastAPI, PySpark, Flask, React, Pandas, NumPy, Pytest

Cloud/DevOps: GCP, Azure, AWS, Docker, Kubernetes, CI/CD, Microsoft Fabric, MLFlow, Firebase

Tools: Git, Bitbucket, Jira, Linux (WSL), Postman, Redis, Flutter, Google Maps API